Guide: Merging Paths of Different Lengths

Collatz proof project — September 5, 2026

What changed

Earlier replacements required the two paths to take the same number of steps. Allowing different lengths finds additional smaller starting seeds that reach the same endpoint. Lean now checks when such a replacement preserves noncontraction for the entire future orbit.

What noncontraction means

After t shortcut steps with j odd steps, the multiplicative coefficient is $3^j/2^t$. A seed is called noncontracting when that coefficient is at least one for every prefix. This is stronger than merely seeing a finite initial run without descent. No infinite noncontracting seed is constructed.

The two checks needed for transport

The smaller seed's path must have coefficient at least one at every step before the meeting. At the meeting, its coefficient must be at least as large as the original path's coefficient. Afterward the two paths follow the same values, so they receive the same further multiplicative factors. Those two checks preserve full noncontraction if the original seed has it.

This matters because shifting a noncontracting seed forward does not automatically give another noncontracting seed. The comparison at the meeting accounts for the coefficient already accumulated before that shift.

A concrete infinite family

For any whole number $Q \geq 0$, take

$$N = 111 + 4374Q, \qquad M = 103 + 4096Q.$$

The larger seed reaches $167 + 6561Q$ in one step. The smaller seed reaches the same value in twelve steps with eight odd steps. Its prefix passes both transport checks. For $Q = 0$, this is the meeting of 111 and 103 at 167. The difference $N - M = 8 + 278Q$ is always positive.

Thus, if N were noncontracting, the smaller M would be noncontracting too. A least positive noncontracting seed cannot be in this progression. This does not show that every member converges, or cover all possible least seeds. The meeting occurs before any forward descent of N.

What the search establishes

Among 7,495 seeds below 2^{18} with noncontracting prefixes through time 18, the previous equal-time/equal-count test covers 1,391 as smaller merges. Searching different lengths through 18 adds 3,768, all with certificates satisfying the new transport theorem. There are still 2,336 unresolved seeds within this search, including 27 and 703.

These counts concern finitely many starting seeds and bounded path lengths. They are exact Python results, not kernel-certified global coverage. An unresolved seed can have a useful longer replacement; it is not a counterexample to Collatz.

Reproduce and continue

From `lean/`, run `lake build Collatz.DominatingMerge`. The search is `analysis/variable_length_surgery.py`, with a matching test file and saved results. The next challenge is to find arithmetic conditions that guarantee a suitable replacement beyond these bounded searches. Neither a proof of nondivergence nor a full Collatz proof has been obtained.